GKS Pre-Connect Hackathon 2026: Report Out

April 13, 2026

RefGetStore

- Identify libraries to incorporate and gaps that needed resolution
  - refgetstore, ferro-hgvs, cdot
- nodejs bindings for refgetstore to be used from Allele Registry
- Investigate ferro-hgvs data loading, and improvements needed
- Loading coding and protein refseq data into public refgetstore bucket
- Loading additional aliases needed by vrs-python into VRS refgetstore bucket

VRS validation expansion

Activities

- Current validation was limited, identified missing variant types based on spec
- Added MNVs, Insertions and Deletions + TODOs list for CopyNumbers
- We were missing alternative digest generator
  - Is being worked on in the other group
  - Already used and improved their code with the new tests
- Discussed need for resource users to onboard their reference/sequences

Notes

- Documentation to follow by Larry
- Serialize statements missing
- Needs stress/volume testing - suggestion to use NIST data
- Needs more thought on haplotypes, fusions, methylation and graph genomes
- Suggestion to merge different issues into one ‘validation’ module

https://github.com/ga4gh/vrs/tree/2.1.0-connect_2026_%2310 -> branch where work was done

Quick Start Guide

Quick Start objectives

Project Details

Proposal: Each product (VRS, Cat-VRS and VA-Spec) should have a Quick Start under Read The Docs, to support the following learning objectives for users who have worked through the content:

1. New users should be aware of the github repos for both the schema (e.g. https://github.com/ga4gh/vrs) and reference implementation (e.g. https://github.com/ga4gh/vrs-python), and should have a basic understanding of the resources available in each repo.
2. Further, new users should have links to notebooks in the reference implementation repo, where those notebooks illustrate the basic operations supported by the reference implementation
3. Also, those users should have links to sample datasets which contain objects (both in YAML and JSON formats) that they can experiment with, along with knowledge gained from the notebooks, for prototyping their own software development efforts that build upon the reference implementation.

Quick Start Guide outputs include:

1. “Quick Start” ReadtheDoc pages added or updated to VRS, VA-Spec, and Cat-VRS
2. “Getting Involved” ReadtheDoc pages added to VRS and VA-Spec
3. Makefile written to compile example yaml files to json and create a README for VRS and Cat-VRS
4. Makefile written to compile validation yaml files to json for VRS
5. Google Colab notebook written for Cat-VRS Quick Start
    a. Runs fully in browser, no installation needed, no coding needed!

Forthcoming Pull Requests for VA-Spec and VRS!

Introduction
How Cat-VRS Works
Quick Start
Getting Involved
Data Model
Implementation Guide
Releases
Appendices

Quick Start

Cat-VRS development consists of two parts, with two interconnected GitHub repos

- https://github.com/ga4gh/cat-vrs is used primarily for schema development.
- https://github.com/ga4gh/cat-vrs-python is used for the development of the reference implementation, in python.

For most new users, the best way to get started with Cat-VRS is to explore the cat-vrs-python notebooks. There, you will find tutorials for assembling variants in VRS, translating between variant namespaces with SeqRepo, and other topics.

Next, many users may want to start prototyping their own software with some sample data. There are a number of example datasets available in both YAML and JSON at https://github.com/ga4gh/cat-vrs/tree/main/examples.

Finally, Cat-VRS is an open-source initiative, driven by community involvement, and we welcome participation from individuals at all levels of skill and experience! To learn about ways to join working group meetings, participate in monthly virutal hackathons, or simply follow the latest news on Cat-VRS, visit Getting Involved for more details.

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Next

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Examples - Variant Representation Specification

This README is automatically generated from the Makefile and an accompanying Python script. Please edit examples in YAML. When ready to compile, run the Makefile to generate both the JSON versions and this README. From this directory:

```
make all
```

Examples by Class

VRS is a collection of data models or concepts that are used together to represent molecular and systemic variation.

Class	Representative examples
Adjacency	ambiguous_linker, invalid_adjacency, precise_linker, revcomp_breakpoint, sequence_homology, simple_breakpoint
Allele	SPDI_contraction, SPDI_expansion
CisPhasedBlock	simple_haplotype
DerivativeMolecule	sv_derivative_molecule
Terminus	terminal_breakend

Cat-VRs Quick Start.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

Cat-VRS Quickstart Notebook

Welcome! This notebook is a lightweight, hands-on introduction to the Cat-VRS Python reference implementation.

By the end, you will know how to:

- install and import Cat-VRS Python
- create `CategoricalVariant` objects
- validate them using Pydantic
- inspect and compare constraints
- search a small in-memory collection of **Categorical Variants (catvars)**
- understand how recipes can define reusable patterns such as gene-level CNVs

Who this notebook is for

This notebook is designed for three kinds of readers:

- people who are new to Cat-VRS and want an intuitive overview
- Python users who want to start creating and validating catvars
- developers who want to skip ahead and inspect the models and helper functions directly

Data Consumption

MaveMD and Quick Start with VA-Spec

Roadmap goals:

1. Represent calibration methods, e.g. from MaveMD. This activity should align with related activity in modeling variant effect predictions.

2. Formalize and align on contributor roles vs. activities, e.g. "modifier role" from CRO as used in MaveMD for "activity type". Is there a more appropriate vocabulary for the activityType field? Review current options and discuss how we would like to make recommendations. May involve a ground-up implementer activity. Consider FHIR RIM prior work. Consider W3C PROV prior work.

3. Address how common VA-spec representation is highly redundant. We should create user guides for how to make these less repetitive in large datasets. Common patterns leveraging cross-document references. Standard profile bundles that leverage these references in a consistent, normalized way. Include discussion on a defaults definition syntax. This should also allow patterns for bundling multiple annotations under one variant.

4. Develop reference tooling for normalized and denormalized flavors of VA-Spec objects.

5. Provide more detailed diagrams about concept relations, e.g. the core model for VA Spec. We should also include illustrative examples for multiple-annotation bundle representations described in goal #3; existing publications and published data documents would be beneficial for this effort. E.g. predictors and functional evidence working together under a clinical variant classification.

https://github.com/ga4gh/gks-portal/issues/70

Scott calibration

BASELINE SCORE -4.00 | *No description provided by author*

Range label	Normal	Loss of function
Classification	Normal	Abnormal
Numeric interval	(−∞, 0.00)	[0.40, ∞)
Evidence strength	BS3_STRONG	PS3_STRONG
OddsPath ratio	0.04	24.90
Variant count	14,973	1,612

SOURCES Thresholds : Scott A et al. (2022) — Method : Brnich SE et al. (2019) — Evidence calcs : Scott A et al. (2022)

MaveMD and Quick Start with VA-Spec

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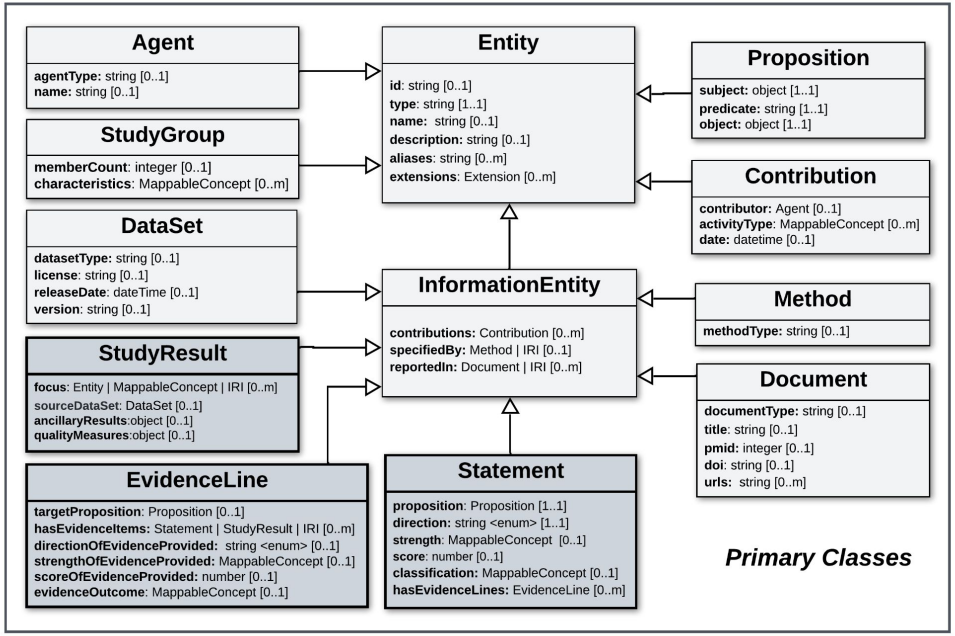
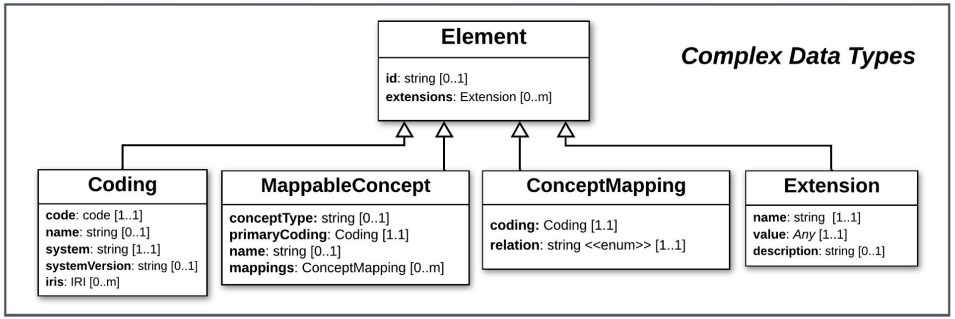
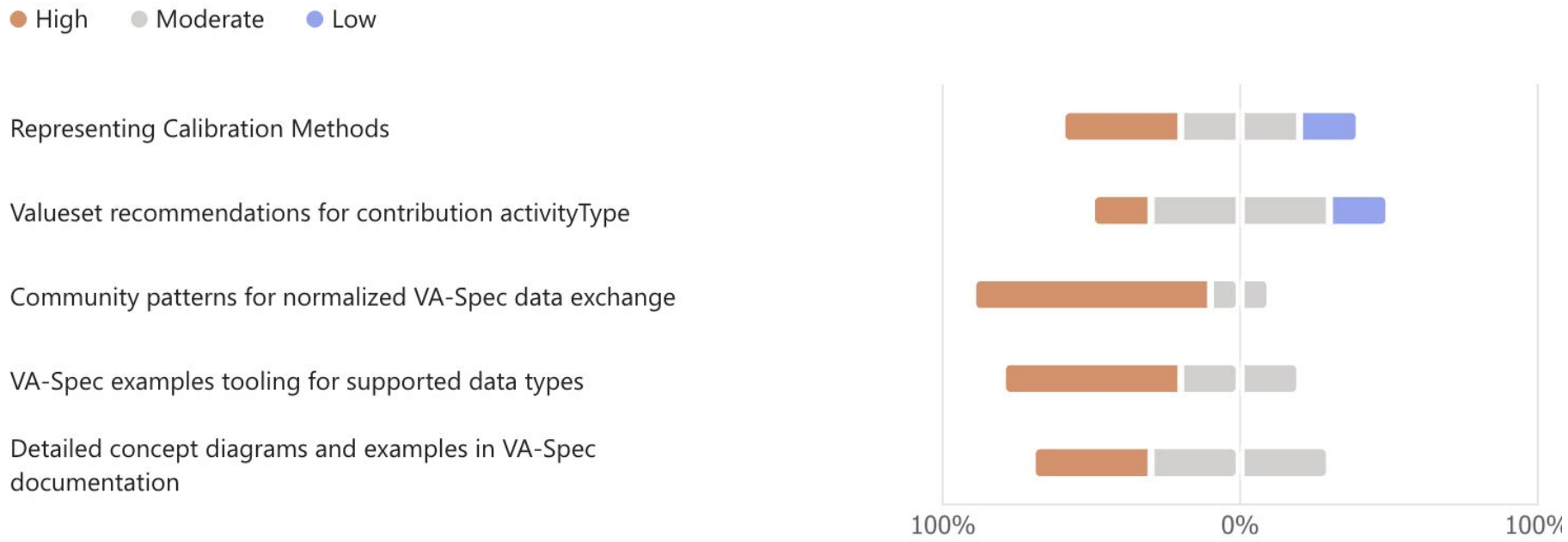

Agent
agentType: string [0..1]
name: string [0..1]
StudyGroup
memberCount: integer [0..1]
characteristics: MappableConcept [0..m]
DataSet
datasetType: string [0..1]
license: string [0..1]
releaseDate: dateTime [0..1]
version: string [0..1]
StudyResult
focus: Entity | MappableConcept | IRI [0..m]
sourceDataSet: DataSet [0..1]
ancillaryResults:object [0..1]
qualityMeasures:object [0..1]
EvidenceLine
targetProposition: Proposition [0..1]
hasEvidenceItems: Statement | StudyResult | IRI [0..m]
directionOfEvidenceProvided: string <enum> [0..1]
strengthOfEvidenceProvided: MappableConcept [0..1]
scoreOfEvidenceProvided: number [0..1]
evidenceOutcome: MappableConcept [0..1]
Entity
id: string [0..1]
type: string [1..1]
name: string [0..1]
description: string [0..1]
aliases: string [0..m]
extensions: Extension [0..m]
InformationEntity
contributions: Contribution [0..m]
specifiedBy: Method | IRI [0..1]
reportedIn: Document | IRI [0..m]
Statement
proposition: Proposition [1..1]
direction: string <enum> [1..1]
strength: MappableConcept [0..1]
score: number [0..1]
classification: MappableConcept [0..1]
hasEvidenceLines: EvidenceLine [0..m]
Proposition
subject: object [1..1]
predicate: string [1..1]
object: object [1..1]
Contribution
contributor: Agent [0..1]
activityType: MappableConcept [0..m]
date: datetime [0..1]
Method
methodType: string [0..1]
Document
documentType: string [0..1]
title: string [0..1]
pmid: integer [0..1]
doi: string [0..1]
urls: string [0..m]
Primary Classes


Element
id: string [0..1]
extensions: Extension [0..m]
Complex Data Types
Coding
code: code [1..1]
name: string [0..1]
system: string [1..1]
systemVersion: string [0..1]
iris: IRI [0..m]
MappableConcept
conceptType: string [0..1]
primaryCoding: Coding [1.1]
name: string [0..1]
mappings: ConceptMapping [0..m]
ConceptMapping
coding: Coding [1.1]
relation: string <<enum>> [1..1]
Extension
name: string [1..1]
value: Any [1..1]
description: string [0..1]

Relative priority

Select your preferred prioritiy level of VA-Spec gaps in producer-consumer data exchange



ahwagner 14 minutes ago · edited by ahwagner Edits Member •••

Goals derived from this session:

1. Refined the ClinVar-GKS dataset aggregate classification approach. Documentation will be updated for the grouping and ranking steps and to align between germline and somatic data representation.
2. Keeping consistent use of propositions in aggregations to limit the number of possible genomic classification types that groups may want to ingest.
3. Nobody in the room wants the aggregated P/LP;Other type aggregate classifications (previously described as L4 aggregates), as this leads to erroneous use of star ratings in downstream analyses.

Noting that we didn't get to any of the questions in this ticket but all came away with a much deeper understanding of the ClinVar resource.

https://github.com/ga4gh/gks-portal/issues/65

AnyVar

- Struggled to setup AnyVar (rip)
- Completed **AnyVar: Implement non-registry translation endpoints**
  - Attempted to deploy AnyVar with these changes
- Identified ways to improve documentation for those contributing to AnyVar
- Made issue to come up with a better name for translation methods (`translate_from` and `translate_to` aren't intuitive)
  - Add subject to the name of methods. For example `translate_VRSfrom` and `translate_VRSto (?)`
- Started implementing method `_to_gnomad` (VRS Object → gnomAD VCF)

GET `/anyvar/translate_from/{fmt}/{identifier}` Translate the specified identifier to a VRS Object

Return the full VRS Object for the given variant identifier

Parameters Cancel

Name	Description
fmt * required string *(path)*	Format for identifier (i.e. hgvs, gnomad) gnomad
identifier * required string *(path)*	Identifier 7-140753336-A-T

Execute | Clear

Responses

Curl

```
curl -X 'GET' \
  'https://services.genomicmedlab.org/anyvar/translate_from/gnomad/7-140753336-A-T' \
  -H 'accept: application/json'
```

Request URL

```
https://services.genomicmedlab.org/anyvar/translate_from/gnomad/7-140753336-A-T
```

Server response

Code	Details
200	Response body


```
{
  "object": {
    "id": "ga4gh:VA.Otc5ovrw906Ack087o1fhegB4jDRqCAe",
    "type": "Allele",
    "digest": "Otc5ovrw906Ack087o1fhegB4jDRqCAe",
    "location": {
      "id": "ga4gh:SL.nhul5x5P_fKjGEpY9PEkMIekJfZaKom2",
      "type": "SequenceLocation",
      "digest": "nhul5x5P_fKjGEpY9PEkMIekJfZaKom2",
      "sequenceReference": {
        "type": "SequenceReference",
        "refgetAccession": "SQ.F-LrLMe1SRpfUZHkQmvkVKFEGaoDeHul"
      },
      "start": 140753335,
      "end": 140753336
    },
    "state": {
      "type": "LiteralSequenceExpression",
      "sequence": "T"
    }
  }
}
```


Download

Gene-Disease Associations

Gene Disease Associations

- Reviewed existing data models of Epic and ClinGen implementations of Gene Validity associations
- Found substantial concordance
- Agreed on common structure for proposition object
  - Subject Gene (by reference)
  - Object Disease (by reference)
  - Qualifier Mode of Inheritance (by reference)
  - Consider name of VariantValidityProposition
  - Posibility for situating the concept in a larger ontological structure
- Would like to move forward codifying this in a GA4GH / GKS context
- Discussed Gene/Sequence Feature model and relationship; expansion of concept to include more precisely defined genetic mechanism.